

Plantarflexor Moment Arm and Muscle Volume Predict Torque-Generating Capacity in Young Men

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Abstract

Muscle volume is known to correlate with maximal joint torque in humans, but the role of muscle moment arm in determining maximal torque is less clear. Moderate correlations have been reported between maximal isometric knee extensor torque and knee extensor moment arm, but no such observations have been made for the ankle joint. It has been suggested that smaller muscle moment arms may enhance force generation at high rates of joint rotation, but this has not yet been observed for ankle muscles *in vivo*. The purpose of the present study was to correlate plantarflexor moment arm and plantarflexor muscle volume with maximal plantarflexor torque measured at different rates of plantarflexion. Magnetic resonance imaging was used to quantify the plantarflexor moment arm and muscle volume of the posterior compartment in twenty healthy young males. Maximal plantarflexor torque was measured isometrically and at three plantarflexion speeds using an isokinetic dynamometer. Plantarflexor torque was significantly correlated with muscle volume ($0.222 < R^2 < 0.322$) and with muscle moment arm at each speed ($0.323 < R^2 < 0.494$). While muscle volume was strongly correlated with body mass and stature, moment arm was not. The slope of the torque-moment arm regression line decreased as the rate of joint rotation increased, indicating that subjects with small moment arms experienced smaller reductions in torque at high speeds. The findings of this study suggest that plantarflexor moment arm is a determinant of joint strength that is at least as important as muscle size.

Introduction

Muscle-tendon units are linear actuators that generate movement only through their ability to produce or resist joint rotation. The joint moment produced by a muscle is the product of the force carried by the muscle's tendon and the moment arm of the muscle-tendon unit about the center of joint rotation. The capacity for joint rotation in an individual human subject thus depends on both muscle force generation and muscle leverage, but the influence of only one of these factors – muscle force – has been investigated thoroughly. There are many published studies documenting correlations between joint strength (*i.e.*, maximum joint torque) and muscle size, which we understand to be a direct determinant of muscle force. Examples of such studies relating strength to size may be found for muscles crossing nearly every major upper and lower extremity joint. At the ankle, for example, interindividual variation in triceps surae muscle volume has been shown to explain 69% to 80% of the variance in maximal plantarflexor torque (11, 21). Very strong correlations have been found between both elbow flexor and extensor muscle volume and their associated maximal joint torques ($R^2 = 0.87$ and $R^2 = 0.85$, respectively) (9).

In contrast to effects of muscle size, the influence of muscle moment arm on variation in strength across individuals has received little attention. We know of only one such study (6), in which a moderate positive correlation ($R^2 = 0.25$) between maximal isometric knee extensor torque and knee extensor moment arm was reported for young healthy subjects. A stronger correlation was found between maximal isometric extensor torque and extensor muscle volume ($R^2 = 0.60$) in the same study. While direct correlations between strength and moment arm have not been reported in other studies, several recent investigations have identified possible links between athletic performance and ankle muscle moment arm. Running has been found to be less energetically

costly for runners with short heels (24, 26), perhaps because the higher muscle forces necessary for running with a short plantarflexor moment arm result in greater elastic energy storage in the Achilles tendon. Other studies have reported comparisons between trained sprinters and height-matched non-sprinters revealing that sprinters have significantly smaller plantarflexor moment arms (5, 16).

A longer Achilles tendon moment arm will, by definition, allow a greater plantarflexor moment to be generated for a given muscle force, but the role of this moment arm in determining ankle joint strength may require consideration of the effects of joint angular velocity as well as mechanical advantage. Examination of human wrist extensors (18) has shown that some muscles are well suited for generating force during rapid joint rotations because they have longer muscle fibers and shorter moment arms, both of which would tend to favor muscle force by reducing sarcomere shortening velocities. Shorter plantarflexor moment arms among human sprinters (5, 16) and observations that the propulsive muscles of animals specialized for sprinting have smaller moment arms relative to the lengths of their associated muscle fascicles (3, 32) also suggest that muscular joint moment and power may be maximized by trading leverage for force generation. Computer simulations using musculoskeletal models have also been used to demonstrate that short plantarflexor moment arms may augment plantarflexor force during rapid plantarflexion by reducing plantarflexor shortening velocity (5, 16, 23), but these simulations did not model adaptations in muscle architecture that may accompany variation in moment arm (14).

The purpose of this study was to use a dynamometer to measure maximal isometric and isokinetic plantarflexor moments in healthy young male subjects to determine if muscular joint strength was correlated with Achilles tendon moment arm at varying rates of plantarflexion.

Achilles tendon moment arm and plantarflexor muscle volume were estimated from magnetic

resonance image data. We hypothesized that maximum plantarflexor moment would be positively correlated with both muscle moment arm and muscle volume during isometric and slow isokinetic contractions. We further hypothesized that the dependence of strength upon moment arm would be reduced during faster isokinetic plantarflexions because subjects with larger plantarflexor moment arms would see greater torque reductions at high speeds due to force-velocity effects.

Materials and Methods

Twenty healthy young adult males participated in this study. The average age of the participants was 26.0 ± 3.5 y and their mean stature and body mass were 177.7 ± 7.7 cm and 76.3 ± 15.6 kg, respectively. All human subjects testing procedures were approved by the Institutional Review Board of The Pennsylvania State University.

Plantarflexor strength was measured with subjects seated in a System 3 isokinetic dynamometer (Biodex Medical Systems, Shirley, NY, USA) with the right foot unshod and secured to the dynamometer foot plate with stiff nylon straps to prevent foot movement. The lateral malleolus was aligned with the dynamometer motor spindle. The right knee was fully extended and the right thigh was strapped to the seat to prevent knee flexion. Neutral ankle position (0°) was defined to occur when a right angle was formed by the foot plate and the long axis of the shank. Dorsiflexion was defined as negative and plantarflexion was defined as positive. To minimize heel motion during testing, the dynamometer seat was moved toward the foot plate until the investigator could not easily advance the seat further. During testing no visible motion of the heel with respect to the footplate was apparent. To set the maximum dorsiflexion angle, subjects' ankles were slowly rotated into dorsiflexion until subjects reported slight plantarflexor tightness;

106 this occurred between 10-13° dorsiflexion for all subjects. The maximum plantarflexion angle
107 was set to 30° for all subjects.

108 Maximal plantarflexor torque was measured under both isometric and isokinetic conditions. In
109 isometric tests, subjects were instructed to push against the footplate maximally for two seconds
110 as isometric plantarflexor torque was measured with the ankle in neutral position. These maximal
111 isometric contractions were performed with rest periods of at least five seconds between each
112 effort. Three maximal isometric contractions were collected for each subject and the best trial
113 was used to define peak isometric torque. During isokinetic tests, maximal plantarflexor
114 contractions were performed as the foot plate was rotated in the plantarflexion direction at 30,
115 120, 210, and 300°·s⁻¹. Subjects used a handheld switch to initiate foot plate rotation manually
116 and were instructed to ‘simultaneously press the hand switch and press as hard and fast as
117 possible against the foot plate’. To permit acclimation to the isokinetic trials, subjects performed
118 at least three practice contractions at each velocity. Subjects then completed a set of five
119 maximal contractions at each of the four plantarflexion speeds in randomized orders. Five
120 seconds of rest was provided within each speed contraction and one minute of rest was provided
121 between conditions. Peak torques did not consistently occur during the first trial of each
122 condition, suggesting that fatigue effects were minimal. During each contraction, subjects
123 received verbal encouragement from the same investigator to plantarflex maximally. The peak
124 torque measured as the ankle passed through the neutral position was analyzed for each
125 condition. Because five subjects were unable to consistently accelerate the footplate to 300°·s⁻¹ at
126 neutral position, this speed was excluded from subsequent analysis.

127 To quantify plantarflexor moment arm and muscle volume, magnetic resonance images of each
128 subject were acquired with a 3.0 T Siemens Trio scanner (Siemens; Erlangen, Germany) while

subjects remained passive. Subjects were positioned supine on the scanner bed with both knees fully extended. The right ankle was placed on an MR-compatible ankle-positioning device (5) that was constructed from plastic and fastened to the scanner bed prior to imaging. This device supported the foot while allowing for manual positioning of the ankle. Stiff straps secured the right foot to the ankle-positioning device to minimize movement during scans. Images were acquired with the ankle positioned at 10° dorsiflexion, neutral ankle position, and 10° plantarflexion (Three dimensional 3D isotropic T1 weighted sequence; TE: 1.31 ms, TR: 3.96 ms, 300 mm field of view, 0.6 mm voxel size). The complete volume of the lower leg was acquired with the ankle in neutral position (3D isotropic T1 weighted; TE: 1.09 ms, TR: 3.96 mm, 500 mm field of view, 0.9 mm voxel size). In addition to MR scanning, B-mode ultrasonography (Aloka 1100; transducer: SSD-625, 7.5 MHz and 39mm scan width; Wallingford, CT, USA) was used to measure the right medial gastrocnemius thickness and pennation angle, from which fascicle length was calculated, using methods previously employed by Lee and Piazza (16). During the ultrasound scans, subjects were standing with the ankle in neutral position.

Magnetic resonance image data were processed using Osirix software (Pixmeo, Geneva, Switzerland). Quasi-sagittal plane images of the ankle were reconstructed and printed onto transparent sheets. The centers of rotation between the tibia and talus at the neutral position (Figure 1) were determined using a modified Reuleaux method that has been described in detail elsewhere (5, 8, 19, 25). Plantarflexor moment arm was defined as the shortest (perpendicular) distance between the midline of the Achilles tendon and the center of ankle rotation. To measure plantarflexor muscle volume, 1 mm thick axial slices of the shank were reconstructed in Osirix viewer and every tenth image was analyzed to obtain 1 cm thick axial slices (10). The margins of

the triceps surae were difficult to identify on some images, so the entire posterior compartment of the leg was outlined on each slice using ImageJ software (version 1.45s; NIH, Bethesda, MD, USA). Total plantarflexor muscle volume was found as the sum of the products of the slice areas and the slice thickness (Figure 2). To assess reliability, estimation of moment arm and muscle volume were repeated for three subjects using image data collected on a second day. The day-to-day differences in moment arm and muscle volume averaged across the three subjects were 2.7 % and 0.8 %, respectively.

Simple linear regressions were performed to test for correlations between torques and either plantarflexor moment arm or muscle volume, and torques were also regressed simultaneously upon plantarflexor moment arm and muscle volume in multiple linear regressions. Moment arm and muscle volume were also regressed against one another and against body mass and stature. A repeated-measures one-way analysis of variance (ANOVA) with post-hoc Bonferroni-corrected pairwise mean comparisons was performed to determine if there were differences in plantarflexor torque across the four velocity conditions (isometric and 30°s^{-1} , 120°s^{-1} , and 210°s^{-1}). To test the hypothesis that moment arm influences ankle torque, a linear mixed model was developed with maximal plantarflexor torque as the outcome variable, moment arm as the predictor variable, and muscle volume as a covariate. This analysis controlled for muscle volume while testing whether the slope of the moment arm-torque regression line at each non-zero speed (30 , 120 , and 210°s^{-1}) differed significantly from the slope found for the isometric regression model. The linear mixed model was performed in the R statistical computing environment (Version 2.15.1, R Development Core Team; www.r-project.org), while ANOVA was performed using IBM SPSS (Version 20, Armonk, NY, USA). The level of significance was set at $\alpha = 0.05$.

Results

Maximal isometric plantarflexor torque was moderately and significantly correlated with both plantarflexor moment arm and plantarflexor muscle volume ($p \leq 0.01$ for both; Table 1; Figures 3 and 4). Maximal plantarflexor torque measured during isokinetic tests was moderately and significantly correlated (all $p \leq 0.036$) with plantarflexor volume at each of the three speeds tested (Table 1; Figure 4). Stronger correlations between maximal isokinetic torque and plantarflexor moment arm were found at each of the speeds tested (all $p \leq 0.002$; Table 1; Figure 3). Multiple regression results showed strong correlations between plantarflexor torque and both moment arm and muscle volume during isometric and isokinetic contractions (all $p \leq 0.007$; Table 1).

Strong positive correlations were found between muscle volume and body mass ($R^2 = 0.757$, $p < 0.001$) and between muscle volume and stature ($R^2 = 0.562$, $p < 0.001$), but the correlations between moment arm and body mass ($R^2 = 0.140$, $p = 0.105$) and between moment arm and stature ($R^2 = 0.156$, $p = 0.085$) were weaker and insignificant. A weak correlation that approached significance was found between muscle volume and moment arm (Figure 5; $R^2 = 0.191$, $p = 0.054$).

ANOVA revealed that plantarflexion velocity had a significant effect ($p < 0.001$) on maximal plantarflexor torque, with torque averaged across subjects decreasing from 169.4 ± 52.9 N m during isometric contractions to 76.1 ± 28.1 N m during isokinetic contractions at $210^\circ \cdot s^{-1}$ (Table 2). Post-hoc mean comparisons of torque between speed conditions showed that each torque was different from every other (all $p < 0.001$).

The linear mixed model (Table 3) revealed that plantarflexor moment arm had a significant effect on plantarflexor torque when controlling for muscle volume ($p = 0.002$). At fast isokinetic contractions of $210^{\circ}\cdot s^{-1}$, a smaller moment arm had a positive effect on torque compared to isometric contractions ($p = 0.048$; Table 3). However, during slower isokinetic contractions of $30^{\circ}\cdot s^{-1}$ and of $120^{\circ}\cdot s^{-1}$, the moment arm's effect on torque did not differ from isometric ($p = 0.272$ and 0.078 , respectively).

Discussion

The results of the present study supported our hypotheses that maximal plantarflexor torque would be positively correlated with both plantarflexor moment arm and plantarflexor muscle volume during both isometric and isokinetic contractions (Figures 3 and 4). While muscle volume was strongly correlated with both body mass and stature, no significant correlation was noted between moment arm and either body mass or stature, suggesting that the influence of moment arm upon plantarflexor torque is more complex than a simple expression of body size. Further, moment arm and muscle volume were not strongly or significantly correlated with one another, and correlation between torque and moment arm was evident even when controlling for muscle volume. The results tended to support our hypothesis that the association between plantarflexor moment arm and plantarflexor torque would weaken as the rate of ankle rotation increased, although the magnitude of plantarflexor torque did not increase in subjects with smaller moment arms. In addition, we found stronger correlations between torque and moment arm during isokinetic trials (Tables 1 and 3). To our knowledge, the present study is the first to establish a correlation between maximal plantarflexor torque and plantarflexor muscle moment arm in humans.

The magnitudes of our strength measurements are similar to values reported previously. The average peak isometric plantarflexor torque of 169 ± 53 N m measured in the present study compares favorably to means of 145 ± 9 N m (20), 120 ± 8 N m (29), and 171 ± 32 N m (22) measured for young male subjects of similar size. As would be expected due to force-velocity effects, the mean isokinetic torques we measured decreased with increasing plantarflexion speed, with mean peak torque of $76 \text{ N m} \pm 28 \text{ N m}$ measured at $210^\circ \cdot \text{s}^{-1}$ (Table 2); comparable reductions in peak isokinetic torques with speed have been reported (29) for subjects of similar size, with mean peak torque of 57 ± 9 N m occurring at $250^\circ \cdot \text{s}^{-1}$.

The magnitudes of the plantarflexor moment arms we measured (5.3 ± 0.6 cm) (Table 2) were comparable to those reported by two previous investigators (8, 25) who used similar methods to measure moment arms in healthy young males. These studies reported moment arms at neutral position of 5.2 ± 0.4 cm (8) and 5.4 ± 0.3 cm (25). During image processing it was difficult to clearly identify the outline of the triceps surae so the posterior compartment of the leg was reported instead, and this compartment includes popliteus, which is not a plantarflexor. Further, this choice excluded the volumes of the plantarflexors fibularis longus and fibularis brevis. The effects of these choices were mitigated somewhat by the fact that these muscles are small relative to the other plantarflexors that comprise the posterior compartment (30, 31). We were able to compare lateral and medial gastrocnemius muscle volumes from our subjects to those reported in previous studies. The mean lateral and medial gastrocnemius volumes of $167.2 \pm 29.8 \text{ cm}^3$ and $304.0 \pm 56.7 \text{ cm}^3$, respectively, that we measured are comparable to values of $140.8 \pm 27.7 \text{ cm}^3$ and $243.7 \pm 33.0 \text{ cm}^3$ measured previously using similar methods (10). The correlation we found between muscle volume for the posterior compartment and maximal isometric torque ($R^2 =$

0.322), however, was weaker than the very strong correlation ($R^2 = 0.80$) previously reported between triceps surae volume and torque in for young male subjects (21).

Our results show a weak and insignificant correlation between muscle volume and moment arm ($R^2 = 0.191$, $p = 0.054$). A much stronger correlation has been reported between elbow extensor muscle cross sectional area and moment arm ($R^2 = 0.645$) (28). In that study, the authors also found a similarly strong correlation between upper arm circumference and elbow extensor moment arm ($R^2 = 0.493$). We found a similar but weaker correlation between calf circumference and moment arm ($R^2 = 0.234$).

The results of the present study raise interesting questions regarding the factors that influence plantarflexor strength. The correlations we found between torque and moment arm were stronger than those found between torque and muscle volume, suggesting that muscle leverage is at least as important a determinant of strength as the volume of the plantarflexor muscles. Large calf muscles are indicators of plantarflexor strength that are plainly evident to the eye, but a large Achilles tendon moment arm is a just as effective, if much less obvious, predictor of strength. Muscle moment arms are likely to be determined by joint structure and tendon paths and for this reason it may be that moment arms are less readily altered than muscle size once skeletal maturity is reached. Plantarflexor moment arm may determine the potential for plantarflexor strength, or the potential gains that may be achieved through participation in a strength training program.

Plantarflexor moment arm may also be a predictor of age-related decline in mobility. Previous work in our laboratory (16) has shown that walking velocity in slower older adults is strongly and positively correlated with plantarflexor moment arm. As elderly subjects lose muscle mass

261 due to age-related sarcopenia, they use an increasing fraction of their available strength when
262 performing activities such as brisk walking that are not commonly regarded as requiring
263 maximal effort (13). As was the case with young subjects performing maximal plantarflexions in
264 the present study, ankle muscle leverage (moment arm) may become a more important
265 determinant of walking velocity in elderly subjects when muscle force is limited (17).

266 Subjects with smaller plantarflexor moment arms exhibited smaller reductions in torque at high
267 rates of joint rotation than did subjects with larger moment arms (Table 3). However, these
268 differences were not as clear as those predicted by computer simulations (5, 16, 23). Previous
269 findings of differences in moment arm between sprinters and non-sprinters (5, 16) and the
270 predictions of musculoskeletal computer simulations (5, 16, 23) led us to hypothesize that a
271 small moment arm might enhance torque production at high rates of plantarflexion. It may be
272 that the findings for our subjects, who were not regular participants in activities like sprinting, do
273 not generalize well to sprinters. Adaptations to training among sprinters may place the muscle at
274 different operating points on the force-length and force-velocity curves (1, 2, 15, 16). A high
275 ratio of muscle fiber length to moment arm, for example, may reduce muscle shortening velocity
276 and enhance force production at faster rates of rotation (18, 23).

277 The design of our experiment did not allow full consideration of the effects of muscular
278 adaptation or force-length effects. Previous investigators have surgically altered the moment
279 arms of animals to elicit acute muscle adaptations (7, 14), but the subjects in this study received
280 no treatment to alter their moment arms. While it is possible that the optimal fiber lengths were
281 adapted to compensate for interindividual differences in moment arm, we found no relationships
282 between medial gastrocnemius fascicle length and either moment arm ($R^2 = 0.003$) or peak
283 torque measured at any of the speeds tested (all $R^2 \leq 0.027$). We attempted to control for muscle

length effects by assessing torque at neutral ankle position, but we cannot know what point this position represents on the force-length curve. Additional analyses revealed that the correlations between the peak torques that occurred at any point in the ankle motion and moment arm ($0.452 \leq R^2 \leq 0.472$) were very similar in strength to correlations between peak torque at neutral position and moment arm ($0.438 \leq R^2 \leq 0.484$). Further, we failed to find correlations between the ankle angles at which peak torques occurred and moment arm (all $R^2 \leq 0.179$).

Certain additional limitations affected our study. Plantarflexor moment arm was determined using a two-dimensional analysis conducted in a quasi-sagittal plane. While this method has been employed by several previous investigators (5, 8, 19, 25), rotations out of the plane of the image could not be quantified and it is possible that more sophisticated three-dimensional determination of moment arm (12, 27) would have yielded different results. Magnetic resonance images were acquired with the ankles unloaded in this study, and moment arms measured with the joint under load are likely to be larger (19) and may have greater physiological relevance. Because of limited scan time, we were only able to quantify plantarflexor moment arm at neutral ankle position. Our experimental design involved correlation of joint torques generated by several muscles with the plantarflexor moment arm of only a subset of those muscles (i.e., the triceps surae, which insert on the Achilles tendon). While we attempted to minimize heel lift during dynamometer tests, we did not attempt to account for this movement using methods such as those proposed by Arampatzis et al. (4), and we cannot state for certain that our torque measurements were not affected by this artifact.

Only about half of the variance in torque could be accounted for in the present study by variation in moment arm and muscle volume. Maximal joint torque should be affected by several factors not considered in this study, including: tendon stiffness, where the plantarflexors were operating

307 on their muscle force-length curves, physiological cross-sectional area, motor unit activation, and
308 muscle composition. We did not ensure that subjects gave maximum effort during the
309 dynamometer trials using electromyography nor did we attempt to assess true muscle strength
310 using twitch interpolation, and these choices may have led to variation in effort that affected our
311 correlations.

312 Further study is needed to understand better the determinants of joint kinetic performance in
313 populations such as elderly adults and persons with movement disorders for whom mobility is
314 impaired by compromised joint function. These studies should take into account other factors,
315 such as tendon dynamics and muscle fiber length patterns, in order to obtain a clearer
316 understanding of the influence of variation in joint structure on joint function.

317

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323 **Disclosures**

324 No conflicts of interest, financial or otherwise, are declared by the authors.

325 **Author Contributions**

326 J.B. and S.P. conception and design of research; J.B. performed experiments; J.B. analyzed data;
327 J.B. and S.P. interpreted results of experiments; J.B. and S.P. prepared figures; J.B. drafted
328 manuscript; J.B. and S.P. edited and revised manuscript; J.B. and S.P. approved final version of
329 manuscript.

330

331 Table 1. *Coefficients of determination (R^2) for correlations between maximal torques and*
 332 *plantarflexor moment arm and volume.*

Plantarflexion Speed	Independent Variable(s)		
	VOL	MA	VOL and MA
Isometric	0.322 (0.009)	0.315 (0.010)	0.445 (0.007)
30°·s ⁻¹	0.226 (0.034)	0.465 (0.001)	0.506 (0.003)
120°·s ⁻¹	0.243 (0.027)	0.438 (0.001)	0.491 (0.003)
210°·s ⁻¹	0.222 (0.036)	0.484 (0.001)	0.520 (0.002)

333 Numbers in parentheses are the p-values for each correlation. VOL, plantarflexor (posterior
 334 compartment) muscle volume; MA, plantarflexor moment arm.
 335

336 Table 2. *Mean moment arm, muscle volume, maximal torques, and medial gastrocnemius*
 337 *characteristics.*

Measure	Value
Plantarflexor moment arm, mm	53.4 ± 5.6
Plantarflexor muscle volume, cm ³	1348 ± 219
Isometric torque, N m	169.4 ± 52.9
Isokinetic torque at 30°·s ⁻¹ , N m	129.6 ± 47.5
Isokinetic torque at 120°·s ⁻¹ , N m	92.0 ± 35.4
Isokinetic torque at 210°·s ⁻¹ , N m	76.1 ± 28.1
MG pennation angle, °	19.5 ± 6.6
MG fascicle length, mm	51.9 ± 5.7

338 MG: medial gastrocnemius. Data are means ± standard deviation.
 339

Table 3. *Results of multiple regression model using muscle volume and moment arm as predictors of torque (R^2) and the linear mixed model demonstrating the effect of moment arm on isokinetic ankle torque*

Plantarflexion Speed	R^2	MA effects on TOR
Isometric	0.445 (0.007)	-
$30^\circ \cdot s^{-1}$	0.506 (0.003)	$\uparrow$ (0.272)
$120^\circ \cdot s^{-1}$	0.491 (0.003)	$\downarrow$ (0.078)
$210^\circ \cdot s^{-1}$	0.520 (0.002)	$\downarrow$ (0.048)

Numbers in parentheses are p-values for the multiple regression model and the effects of moment arm on isokinetic ankle torque, respectively. Directional arrows in column showing the effects of moment arm (MA) on ankle torque (TOR) indicate whether slope of regression line increased or decreased with respect to isometric torque regression.

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Figure Captions

Figure 1 Magnetic resonance image of a sagittal cross section of the foot and ankle showing the center of ankle rotation and plantarflexor moment arm.

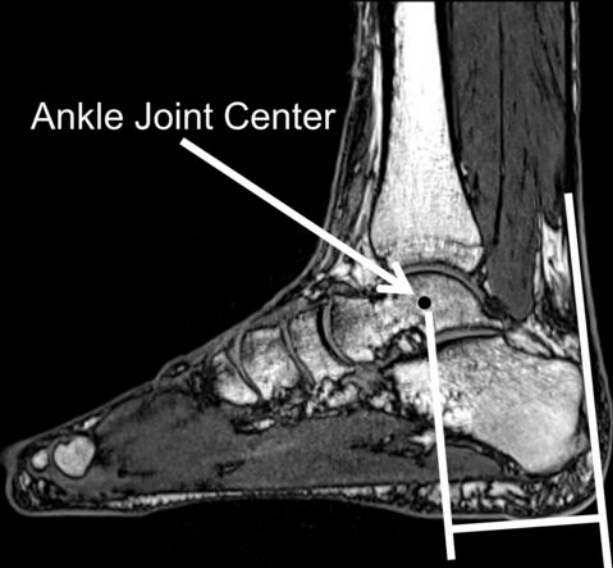
Figure 2 Magnetic resonance image of a cross section of the lower leg showing the area of the posterior compartment filled in with a striped pattern.

Figure 3 Maximal plantarflexor torque measured at neutral position plotted against plantarflexor moment arm for isometric trials (*top*) and the $210^{\circ} \text{ s}^{-1}$ isokinetic trials (*bottom*). Significant correlations were found between torque and moment arm for both conditions ($R^2 = 0.315$; $p = 0.01$ and $R^2 = 0.484$; $p = 0.001$, respectively).

Figure 4 Maximal plantarflexor torque measured at neutral position plotted against posterior compartment muscle volume for isometric trials (*top*) and the $210^{\circ} \text{ s}^{-1}$ isokinetic trials (*bottom*). Significant correlations were found between torque and muscle volume for both conditions ($R^2 = 0.322$; $p = 0.009$ and $R^2 = 0.222$; $p = 0.036$, respectively).

Figure 5 Posterior compartment muscle volume plotted against plantarflexor moment arm for all subjects. A correlation was found between muscle volume and moment arm that nearly reached the level of significance ($R^2 = 0.191$; $p = 0.054$).

Ankle Joint Center



Moment Arm

